

Inch to mm conversion chart: fractions, common sizes, mm to inches

2. Inch fractions in millimetres, 1/64 to 1 inch

Exact values, not rounded (1/64 inch = 0.396875 mm). Sixteenth fractions, the most common on drawings and tools, are in bold.

Inch (fraction)	Decimal inch	mm
1/64	0.015625	0.396875
1/32	0.03125	0.79375
3/64	0.046875	1.190625
1/16	0.0625	1.5875
5/64	0.078125	1.984375
3/32	0.09375	2.38125
7/64	0.109375	2.778125
1/8	0.125	3.175
9/64	0.140625	3.571875
5/32	0.15625	3.96875
11/64	0.171875	4.365625
3/16	0.1875	4.7625
13/64	0.203125	5.159375
7/32	0.21875	5.55625
15/64	0.234375	5.953125
1/4	0.25	6.35
17/64	0.265625	6.746875
9/32	0.28125	7.14375
19/64	0.296875	7.540625
5/16	0.3125	7.9375
21/64	0.328125	8.334375
11/32	0.34375	8.73125
23/64	0.359375	9.128125
3/8	0.375	9.525
25/64	0.390625	9.921875
13/32	0.40625	10.31875
27/64	0.421875	10.715625
7/16	0.4375	11.1125
29/64	0.453125	11.509375
15/32	0.46875	11.90625
31/64	0.484375	12.303125
1/2	0.5	12.7
33/64	0.515625	13.096875

Inch (fraction)	Decimal inch	mm
17/32	0.53125	13.49375
35/64	0.546875	13.890625
9/16	0.5625	14.2875
37/64	0.578125	14.684375
19/32	0.59375	15.08125
39/64	0.609375	15.478125
5/8	0.625	15.875
41/64	0.640625	16.271875
21/32	0.65625	16.66875
43/64	0.671875	17.065625
11/16	0.6875	17.4625
45/64	0.703125	17.859375
23/32	0.71875	18.25625
47/64	0.734375	18.653125
3/4	0.75	19.05
49/64	0.765625	19.446875
25/32	0.78125	19.84375
51/64	0.796875	20.240625
13/16	0.8125	20.6375
53/64	0.828125	21.034375
27/32	0.84375	21.43125
55/64	0.859375	21.828125
7/8	0.875	22.225
57/64	0.890625	22.621875
29/32	0.90625	23.01875
59/64	0.921875	23.415625
15/16	0.9375	23.8125
61/64	0.953125	24.209375
31/32	0.96875	24.60625
63/64	0.984375	25.003125
1	1	25.4

3. Whole inches and common sizes

The foot is 12 inches (304.8 mm) and the yard 36 inches (914.4 mm), also exact values.

Inches	Decimal inch	mm
1	1	25.4
1 1/4	1.25	31.75
1 1/2	1.5	38.1
1 3/4	1.75	44.45
2	2	50.8

Inches	Decimal inch	mm
2 1/2	2.5	63.5
3	3	76.2
3 1/2	3.5	88.9
4	4	101.6
5	5	127
6	6	152.4
8	8	203.2
10	10	254
12 (1 ft)	12	304.8
24 (2 ft)	24	609.6
36 (1 yd)	36	914.4

4. Millimetres to inches, with the nearest fraction

Inches rounded to four decimals. The last column gives the deviation between the nearest 1/64 fraction and the millimetre dimension: at most ± 0.198 mm, half of a 1/64.

mm	Inches	Nearest fraction	Deviation (mm)
1	0.0394	3/64	+0.191
2	0.0787	5/64	−0.016
3	0.1181	1/8	+0.175
4	0.1575	5/32	−0.031
5	0.1969	13/64	+0.159
6	0.2362	15/64	−0.047
8	0.3150	5/16	−0.062
10	0.3937	25/64	−0.078
12	0.4724	15/32	−0.094
15	0.5906	19/32	+0.081
16	0.6299	5/8	−0.125
20	0.7874	25/32	−0.156
25	0.9843	63/64	+0.003
30	1.1811	1 3/16	+0.162
40	1.5748	1 37/64	+0.084
50	1.9685	1 31/32	+0.006
60	2.3622	2 23/64	−0.072
75	2.9528	2 61/64	+0.009
80	3.1496	3 5/32	+0.169
100	3.9370	3 15/16	+0.012
150	5.9055	5 29/32	+0.019
200	7.8740	7 7/8	+0.025
250	9.8425	9 27/32	+0.031
300	11.8110	11 13/16	+0.038

mm	Inches	Nearest fraction	Deviation (mm)
500	19.6850	19 11/16	+0.062
1000	39.3701	39 3/8	+0.125

Reference values taken from the article <https://www.baouch.fr/en/blog/conversion-pouce-mm-tableau/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.